

8

B

$$E_k = \frac{1}{2}m(v_x^2 + v_y^2) \text{ where } v_x = u_x \text{ and } v_y = u_y - gt$$

$$E_k = \frac{1}{2}m(u_x^2 + (u_y - gt)^2)$$

$$E_p = mgh = mg(u_y t - \frac{1}{2}gt^2)$$

From the above equations, the variation with time of the energies is a quadratic relationship.

By the conservation of energy, the increase in G.P.E. is equal to the decrease in K.E. as the ball moves upwards to its highest point and the decrease in G.P.E. is equal to the increase in K.E. as the ball moves downwards back to the ground.

At $t = 0$ and when ball is back to the ground,

$$E_k = \frac{1}{2}m(u_x^2 + u_y^2) \text{ and } E_p = 0$$

At maximum height, $v_y = 0$, $t = \frac{u_y}{g}$

$$E_k = \frac{1}{2}mu_x^2, E_p = mgh = mg\left(\frac{u_y^2}{2g}\right) = \frac{1}{2}mu_y^2$$